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$PbTeB_4O_9$: a lead tellurium borate with unprecedented fundamental building block $[B_4O_{10}]$ and large birefringence†

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Herein, the first lead tellurium borate, $PbTeB_4O_9$, with an unprecedented fundamental building block $[B_4O_{10}]$ was successfully synthesized. The near-parallel alignment of $[B_4O_{10}]$ groups and $[TeO_3]$ polyhedra resulted in a high birefringence (0.099@1064 nm). The structure–property relationship was discussed by using the first-principles calculations.

As an essential optical material, birefringent crystals can be processed into various optical devices owing to their ability to modulate the polarization state of lasers, including photoelectric modulators, optical isolators, and prism polarizers, which are used extensively in various advanced optical systems.[1,2] During the past several decades, some birefringent materials were synthesized and developed, including YVO_4, TiO_2, $LiNbO_3$, $CaCO_3$, MgF_2, and α-BaB_2O_4 (α-BBO) crystals.[3] Even so, scientists have not stopped searching for new birefringent materials, and they have struggled to find excellent birefringent functional materials to enhance crystal optical anisotropy.

Generally, the introduction of special cations could contribute to the birefringence. Emerging examples include $PbB_5O_7F_3$, SbB_5O_9, α-$Ba_3Sc_2(BO_3)_4$, $LaBO_3$, *etc.*[4] Recently, $Sn_2B_5O_9Cl$ was successfully designed and synthesized by Pan's group in a closed system.[5] The introduction of Sn^{2+} cation makes its calculated birefringence about 16 times that of its isostructural $Ba_2B_5O_9Cl$. It was proven that the stereo-chemically Sn^{2+} could greatly increase birefringence, and several Sn^{2+}-containing crystals were also discovered.[6] In addition, Te^{4+} with the lone pair also attracted much interest in designing crystals with considerable birefringence. For example, by introducing Te^{4+} into phosphate, a novel phosphate crystal, $Te_3O_3(PO_4)_2$, with an unexpectedly enhanced birefringence of 0.148@1064 nm was obtained.[7] In addition, the compounds, $K_2(TeO)P_2O_7$, $Rb_2(TeO)P_2O_7$, and $Cs_2(TeO)P_2O_7$, also show quite large birefringence of 0.070, 0.090, and 0.060 at 1064 nm, respectively.[8] Similar to phosphates, Te^{4+} also improves the optical anisotropy of sulfates, allowing them to exhibit large birefringence, for example, the value of $Y_3(TeO_3)_2(SO_4)_2(OH)(H_2O)$ and $Y_2(Te_4O_{10})(SO_4)$ was calculated to be 0.080 and 0.124 at 1064 nm, respectively.[9]

Except for the special cations, the anion framework also has a significant effect on the birefringence. Among them, borates have long been the focus of scientists to explore new birefringent materials because of their rich structures.[10] Generally, boron can be bonded to oxygen atoms through sp, sp^2, and sp^3 hybridization to form $[BO_2]$, $[BO_3]$, and $[BO_4]$ units, which can be further aggregated into many complex structures containing various fundamental building blocks (FBB), such as $[B_3O_7]$ and $[B_3O_6]$ with the ability to produce a large birefringence.[3f,11] Furthermore, to enrich the structural chemistry and enhance the optical performance of borates, fluorination strategies were proposed to improve the polarization anisotropy and hyperpolarizability of $[BO_4]$.[12] Some novel FBBs with B–F bonds were discovered, generating dozens of new fluorooxoborates with excellent birefringent properties.[13] For instance, the famous commercial LiB_3O_5 with FBBs-$[B_3O_7]$ exhibits a perfect nonlinear optical property and large band gap, however, the arrangement of its FBBs is not conducive to generating a large birefringence.[14] Using the famous LiB_3O_5 ($Li_2B_6O_{10}$) as the parent structure, $A_2B_6O_9F_2$ (A = Li, Na, K, Rb, NH_4) series with enhanced birefringence were designed and synthesized.[12a,15] Notably, their FBBs are $[B_6O_{11}F_2]$, which are quite different in the overall structure and optical anisotropy due to the various configurations and composition of their FBBs. Herein, to

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† Electronic supplementary information (ESI) available: Experimental sections, computational methods, crystallographic data tables, powder XRD patterns, optical property measurements for $PbTeB_4O_9$. CCDC 2302636 for $PbTeB_4O_9$. For ESI and crystallographic data in CIF or other electronic format see DOI: https://doi.org/10.1039/d3cc05382b

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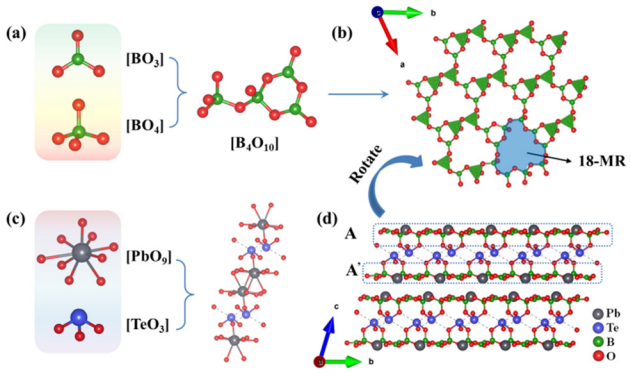

Fig. 1 The crystal structure of $PbTeB_4O_9$. (a) The fundamental building blocks (FBB), $[B_4O_{10}]$ group; (b) the 2D infinite $[B_4O_8]_\infty$ layer with 18-membered rings consisting of $[B_4O_{10}]$ FBBs; (c) the coordination of Pb^{2+} cation and Te^{4+} cation; (d) the 3D framework of $PbTeB_4O_9$.

optimize the arrangement of $[B_3O_7]$, we introduced $[BO_4]$ and lone pair electron cation to make $[B_3O_7]$ tend to be coplanar. After much effort, we designed and synthesized a new borate, $PbTeB_4O_9$, with large birefringence *via* a spontaneous crystallization method in an open system. As far as we know, $PbTeB_4O_9$ possesses the unprecedented FBB-$[B_4O_{10}]$ groups, which further connect mutually to form two-dimensional (2D) infinite $[B_4O_8]_\infty$ layers. The incorporation effect of the nearly complanate $[B_4O_8]_\infty$ layers and profitable arrangement of $[TeO_3]$ polyhedra results in a large birefringence (Δn = 0.099@1064 nm) of $PbTeB_4O_9$.

The structure of $PbTeB_4O_9$ was determined by single-crystal XRD, with the structural data given in Tables S1–S3 in the ESI.† $PbTeB_4O_9$ crystallizes in a triclinic crystal system with the space group, $P\bar{1}$. It features a three-dimensional (3D) framework consisting of $[BO_3]$, $[BO_4]$, $[TeO_3]$, and $[PbO_9]$ polyhedra (Fig. 1). The asymmetric unit of $PbTeB_4O_9$ contains one Pb, one Te, four B, and nine O atoms (Table S2, ESI†). In the structure, B atoms have two types of coordination, *i.e.*, $[BO_3]$ and $[BO_4]$ units, with the B–O distances ranging from 1.341 to 1.399 Å and 1.454 to 1.499 Å, respectively. One $[BO_4]$ and two $[BO_3]$ units constitute a $[B_3O_7]$ ring which further links with another $[BO_4]$ unit forming the FBB, $[B_4O_{10}]$ group (Fig. 1a). The $[B_4O_{10}]$ FBBs expand in the *ab* plane forming 2D infinite $[B_4O_8]_\infty$ layer (A layer) with 18-membered rings (MR) (Fig. 1b). The A layer rotates along the *b* axis, forming A' layer and then the layers stack along the *c* axis in a packing mode of AA′···AA′···AA′. The $[TeO_3]$ polyhedra are located between the A and A′ layers, while $[PbO_9]$ polyhedra are located in the 18-MRs of the $[B_4O_8]_\infty$ layers, which are further connected by weak Te–O bonds (secondary bonds) and Pb–O bonds to construct a final 3D structure (Fig. 1c and d). The Pb atom bonds to nine O atoms forming a $[PbO_9]$ polyhedron with Pb–O bond lengths of 2.346 to 3.098 Å. The $[PbO_9]$ polyhedra connect mutually forming chains along the *b* axis by sharing O atoms and further arranged in order along the *a* and *c* axis (Fig. S1a and S1b, ESI†). A Te atom bonds to three O atoms forming a $[TeO_3]$ polyhedron with the Te–O bond distances of 1.880 to 1.919 Å. The $[TeO_3]$ polyhedra connect mutually by the secondary bond distances of 2.604 to 2.822 Å forming chains along the *b* axis and further arranged in order along the *a* and *c* axis (Fig. S1c and S1d, ESI†).

Notably, the $[B_4O_{10}]$ group is unwonted, for which we summarized the $[B_4O_{10}]$ groups from the ICSD. To the best of our knowledge, the $[B_4O_{10}]$ group consisting of two $[BO_4]$ and two $[BO_3]$ units in $PbTeB_4O_9$ was first reported as a new FBB in borate. As shown in Fig. S2 (ESI†), there are four types of $[B_4O_{10}]$ groups as FBBs in borates. These FBBs are described as follows according to Burns, Christ, and Clark: (4 : 4T), (4 : 3Δ + T), (4 : 2Δ + 2T), where Δ and T denote the $[BO_3]$ triangle and $[BO_4]$ tetrahedron, respectively.[16] We can see that the $[B_4O_{10}]$ group in $PbTeB_4O_9$ is unique. In $PbTeB_4O_9$, the $[B_4O_{10}]$ FBBs expand in the *ab* plane forming a 2D infinite $[B_4O_8]_\infty$ layer, in which $[BO_3]$ units align almost in the same direction. According to Chen *et al.*, such an arrangement promotes a large optical anisotropy.[17] A similar layer can be found in the compound NaB_4O_6F.[18] As shown in Fig. S3 (ESI†), three $[BO_3]$ and one $[BO_3F]$ are connected to the $[B_4O_8F]$ FBB in NaB_4O_6F, which further generate a 2D infinite $[B_4O_6F]_\infty$ layer. The B–O/F layers in NaB_4O_6F are nearly parallel, which results in a large birefringence for NaB_4O_6F (Δn = 0.12@1064 nm). $PbTeB_4O_9$ inherits the excellent layer structure of NaB_4O_6F, which is beneficial to generating a large birefringence.

Interestingly, the 2D infinite $[B_4O_8]_\infty$ layer can also be found in compound $Rb_3B_7O_{12}$.[19] As shown in Fig. 2, $Rb_3B_7O_{12}$ has a 2D B–O framework consisting of three types of $[B_4O_8]_\infty$ layers, which can be divided into A, B, and C layers, with $[BO_3]$ units as linkers located between the layers. The $[TeO_3]$ polyhedra in $PbTeB_4O_9$ replace the $[BO_3]$ linkers in $Rb_3B_7O_{12}$ and work as scissors to cut the connection between the layers owing to the stereo-chemically active lone pairs (SCALP) effect of Te^{4+} cations. Compared with $[B_4O_8]_\infty$ layers in $Rb_3B_7O_{12}$, we can see in Fig. 2c and d that the $[B_4O_8]_\infty$ layers in $PbTeB_4O_9$ tend to be flatter. The strong asymmetric electron distribution of Te^{4+} cations also offers large optical anisotropy. Therefore, the synergistic effect of Te^{4+} cations and $[B_4O_8]_\infty$ layers makes $PbTeB_4O_9$ exhibit a large birefringence, which is ∼2.5 times that of $Rb_3B_7O_{12}$.

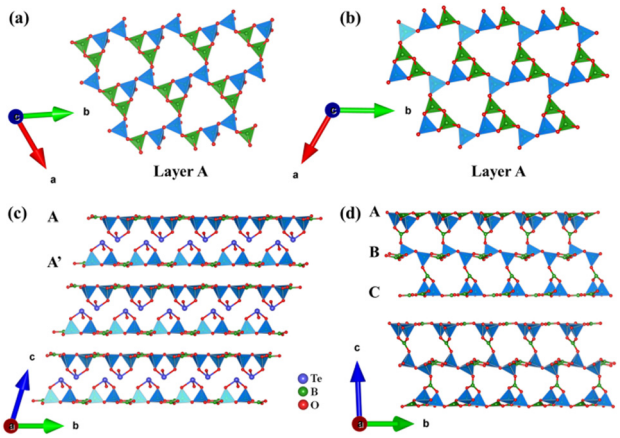

Fig. 2 (a) Layer A of $PbTeB_4O_9$; (b) layer A of $Rb_3B_7O_{12}$; (c) structure of $PbTeB_4O_9$; (d) structure of $Rb_3B_7O_{12}$. (Blue tetrahedra are $[BO_4]$ units and green triangles are $[BO_3]$ units.)

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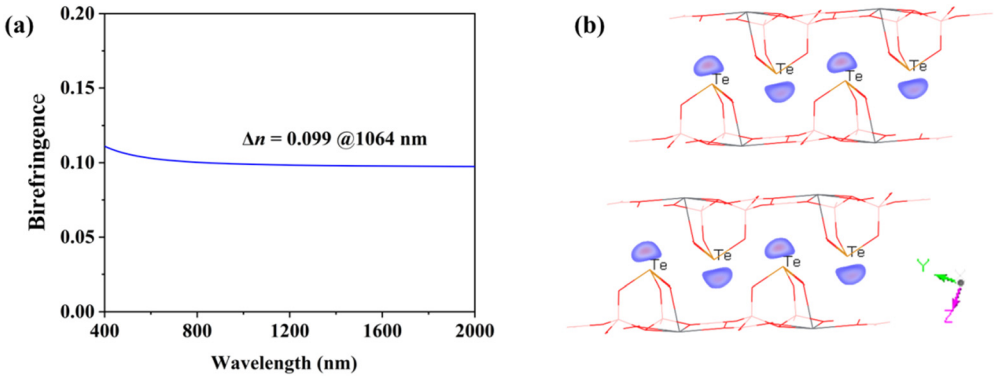

Fig. 3 (a) The calculated birefringence of $PbTeB_4O_9$. (b) Electron localization function diagram of Te atoms.

Polycrystalline samples of $PbTeB_4O_9$ were prepared at 520 °C by conventional solid-state reactions. The powder X-ray diffraction (XRD) pattern of the samples showed good agreement with the calculated one but with several peaks of PbB_4O_7 (Fig. S4, ESI†). The thermal behavior of $PbTeB_4O_9$ is presented in Fig. S5 (ESI†). There are two sharp endothermic peaks around 571 and 590 °C, respectively, on the heating curve. To understand its melting behavior, polycrystalline samples of $PbTeB_4O_9$ were annealed at 580 °C for 24 h for powder XRD inspection. We can see that the samples were decomposed to PbB_4O_7 and TeO_2 at 580 °C, indicating that $PbTeB_4O_9$ is an incongruent melting compound. The IR spectrum of $PbTeB_4O_9$ was carried out verifying the existence of the [BO_3], [BO_4], and [TeO_3] units (Fig. S6 and Table S4 in the ESI†). The energy-dispersive X-ray spectroscopy further verified the presence of Pb, Te, B, and O elements (Fig. S7, ESI†). The UV-Vis-NIR diffuse reflectance spectrum of $PbTeB_4O_9$ was measured, indicating that the UV cut-off edge of $PbTeB_4O_9$ is 242 nm (Fig. S8, ESI†), which is comparable with $Rb_3BaTeB_7O_{15}$ (240 nm) and $LaTeBO_5$ (246 nm), *etc.*[20] The reflectance spectrum was converted to the absorption spectrum with the Kubelka–Munk function,[21] obtaining an experimental band gap for $PbTeB_4O_9$ of about 4.58 eV.

To further explore the relationship between electronic structure and optical properties of $PbTeB_4O_9$, we used first-principles analysis. The optical band gap of $PbTeB_4O_9$ was calculated by the GGA method. As shown in Fig. S9a (ESI†), the valence band (VB) maximum and the conduction band (CB) minimum were not located at the same high symmetry points of the Brillouin zone, resulting in an indirect band gap of 4.27 eV. The optical properties mainly originate from the electronic transitions between the top of the VBs and the bottom of the CBs. As shown in the partial density of states (PDOS) curves of the respective elements in $PbTeB_4O_9$ (Fig. S9b, ESI†), the top of the VBs is dominated by O 2p orbitals, while the bottom of the CBs is mainly occupied by B 2p, Te 5p, and Pb 6p orbitals. It indicates that the linear optical properties of $PbTeB_4O_9$ primarily arise from the [TeO_3] polyhedra, [PbO_9] polyhedra, and the [B–O] groups.

In addition, the obtained birefringence curve is shown in Fig. 3a. The calculated birefringence is about 0.099 at 1064 nm, which is larger than or comparable to those of other reported tellurium borates (Table S5, ESI†). From Table S5 (ESI†), we can see that among the tellurium borates reported so far, the coordination of Te^{6+} is relatively fixed forming the [TeO_6] group with O atoms. Most of the [TeO_6] groups form 0D [Te–B–O] clusters with [BO_3] groups in tellurate borates, and only $Ba_4B_8TeO_{19}$ has a 3D [Te–B–O] framework formed by [B_4O_{11}] and [TeO_6] groups.[22] As the Te^{6+} cations have no SCALP and the B–O framework is poorly aligned, the birefringence of the above compounds is smaller than that of $PbTeB_4O_9$. Whereas in tellurite borates, Te^{4+} adopts flexible coordination geometries, such as [TeO_4], [TeO_3], *etc.* Owing to the SCALP of Te^{4+}, the ideal arrangement of [TeO_4] in TeB_4O_8 leads to its large birefringence.[23] In addition, the contribution of [B–O] groups should not be neglected, such as $LaTeBO_5$.[20a] In short, $PbTeB_4O_9$ exhibits a large birefringence due to the synergistic effect of the parallel arrangement of the [TeO_3] polyhedra and the [B–O] groups. The calculation results of the electron localization function can also support the above statement (Fig. 3b and Fig. S10, ESI†). The high electron density near Te atoms proves the existence of lone-pair electrons. In comparison, Pb atoms have a weak lone pair electron effect. Therefore, the SCALP-containing [TeO_3] polyhedra are favorable structural units for birefringence. Further, the real-space atom-cutting method was used to investigate the real contribution of the three units on the birefringence and the results are shown in Table S6 (ESI†).[24] We can see that the synergistic effect of [Te–O] (~56%) and [B–O] (~37%) groups leads to the large birefringence of $PbTeB_4O_9$.

In conclusion, a new lead tellurium borate, $PbTeB_4O_9$, was successfully obtained *via* a high-temperature solution method in the open air. $PbTeB_4O_9$ has a 3D framework with unprecedented 2D infinite $[TeB_4O_9]_\infty$ layers consisting of novel [B_4O_{10}] FBBs and the SCALP-containing [TeO_3] polyhedra. The diffuse reflectance spectrum shows that $PbTeB_4O_9$ has a wide transmission range with a UV cut-off edge of 242 nm. In addition, the calculated birefringence of $PbTeB_4O_9$ is ~0.099 at 1064 nm. The real-space atom cutting method indicates that the synergistic effect of [TeO_3] polyhedra and [B–O] groups leads to a large birefringence of $PbTeB_4O_9$.

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Conflicts of interest

There are no conflicts to declare.

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